

ENGINEERING
ADMISSIONS ASSESSMENT

D564/11

2023

60 minutes

SECTION 1

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open this question paper until you are told that you may do so. This paper is Section 1 of 2.

A separate answer sheet is provided for this paper. Please check you have one. You also require a soft pencil and an eraser.

Please complete the answer sheet with your candidate number, centre number, date of birth, and name.

At the end of 60 minutes, your supervisor will collect this question paper and answer sheet before giving out Section 2.

This paper contains **two** parts, **A** and **B**, and you should attempt **both** parts.

Part A Mathematics and Physics (20 questions)

Part B Advanced Mathematics and Advanced Physics (20 questions)

You are **strongly** advised to divide your time equally between the two parts: 30 minutes on **Part A** and 30 minutes on **Part B**. The scores for Part A and Part B are reported separately.

This paper contains 40 multiple-choice questions. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt **all** 40 questions. Each question is worth one mark.

For each question, choose the **one** option you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

You **must** complete the answer sheet within the time limit.

You can use the question paper for rough working, but **no extra paper** is allowed. Only your responses on the answer sheet will be marked.

Dictionaries and calculators are NOT permitted.

Please wait to be told you may begin before turning this page.

This question paper consists of 31 printed pages and 5 blank pages.

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PART A Mathematics and Physics

- 1 The surface area of a solid sphere of radius R is equal to the total surface area of 10 solid closed cylinders of radius r and height $4r$.

Which of the following is an expression for R in terms of r ?

(The surface area of a sphere of radius R is $4\pi R^2$.)

A $R = 5r$

B $R = 12r$

C $R = 2\sqrt{5}r$

D $R = \frac{1}{2}\sqrt{10}r$

E $R = \sqrt{10}r$

F $R = \frac{3}{2}\sqrt{10}r$

G $R = \sqrt{15}r$

- 2 A spaceship of mass 10 000 kg is moving at 2.0 m s^{-1} relative to a space station.

The spaceship is captured by a robotic arm attached to the space station and brought to rest by a force of 1000 N.

How far will the spaceship move in its initial direction relative to the space station while the force is being applied?

(Assume that the acceleration of the space station is negligible.)

A 0.050 m

B 0.10 m

C 0.20 m

D 5.0 m

E 10 m

F 20 m

- 3 Which of the following is a correct rearrangement of

$$y = p - \frac{q-r}{s-x}$$

to make x the subject?

A $x = s - \frac{q-r}{p+y}$

B $x = \frac{q-r}{p+y} - s$

C $x = s - \frac{q-r}{p-y}$

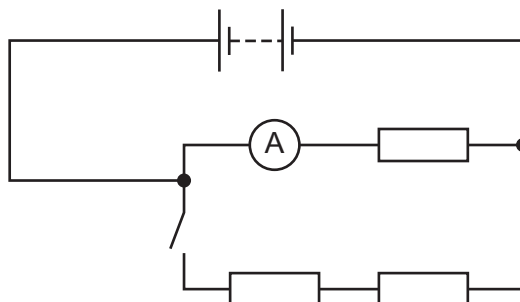
D $x = \frac{q-r}{p-y} - s$

E $x = s - \frac{q-r}{y-p}$

F $x = \frac{q-r}{y-p} - s$

- 4 A circuit is set up as shown. All three resistors are identical.

When the switch is open, the reading on the ammeter is 1.0 A and the power transferred from the battery is 1.0 W .

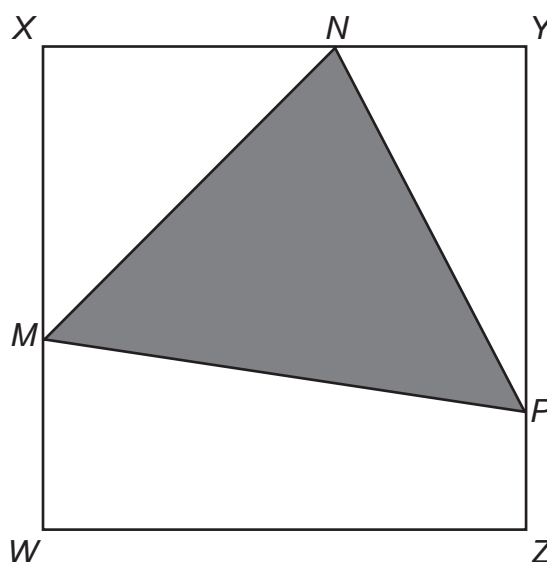


The switch is now closed.

What is the new reading on the ammeter and what is the new power transferred from the battery?

	<i>ammeter reading / A</i>	<i>power transferred / W</i>
A	0.67	0.67
B	0.67	1.3
C	0.67	1.5
D	0.67	2.0
E	1.0	1.0
F	1.0	1.5
G	1.0	2.0
H	1.0	3.0

5



[diagram not to scale]

$WXYZ$ is a square of side length 1.

$WM:MX = 1:2$

$XN:NY = 3:1$

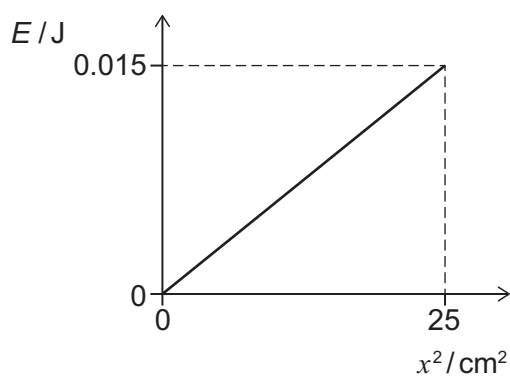
$YP:PZ = 4:1$

What is the area of triangle MNP ?

- A $\frac{1}{3}$
- B $\frac{2}{5}$
- C $\frac{9}{20}$
- D $\frac{1}{30}$
- E $\frac{19}{60}$
- F $\frac{23}{60}$

- 6 A spring is initially unstretched. A force F is used to stretch the spring. The extension x and the energy E stored in the stretched spring are measured for different values of F .

The graph shows how the energy E , in J, varies with the extension squared, x^2 , in cm^2 .



What is the magnitude of F when the spring stores 0.015 J of energy?

- A 0.30 N
- B 0.60 N
- C 1.2 N
- D 1.5 N
- E 2.4 N
- F 3.0 N
- G 30 N
- H 60 N

- 7 Given that

$$\frac{27^{2(x-2)}}{9^{(2x-3)}} = (81)^{\frac{3}{2}}$$

what is the value of x ?

- A 0
- B 2.5
- C 3
- D 6
- E 7.5
- F 9
- G 10.5
- H 12

- 8 A solid, cylindrical metal bar has a uniform cross-sectional area of 12 cm^2 and a volume of 180 cm^3 .

The bar rests on a horizontal surface on one of its circular faces.

The pressure on the surface due to the bar is 0.45 N cm^{-2} .

What is the density of the metal, in g cm^{-3} ?

(gravitational field strength = 10 N kg^{-1})

- A 2.5 g cm^{-3}
- B 3.0 g cm^{-3}
- C 3.75 g cm^{-3}
- D 7.5 g cm^{-3}
- E 15 g cm^{-3}
- F 33 g cm^{-3}

- 9** Last year, the salary of the coach of a football club was 80% of the salary of the star player.

At the start of the new year, the coach received a 15% increase in salary and the star player received a 38% increase in salary.

What percentage of the star player's new salary is the coach's new salary?

- A** 46%
- B** 57%
- C** $61\frac{3}{5}\%$
- D** $66\frac{2}{3}\%$
- E** 77%
- F** $83\frac{1}{3}\%$

- 10** Two samples of pure radioactive isotopes X and Y decay with half-lives of 2 days and 3 days, respectively.

Both X and Y decay in a single step into different stable isotopes.

Initially the number of atoms of X is twice the number of atoms of Y.

After how many days are the expected numbers of atoms of X and Y equal to each other?

- A** The expected numbers of atoms of X and Y are never equal.
- B** 2 days
- C** 3 days
- D** 4 days
- E** 6 days
- F** 12 days

- 11** An athlete's training session consists of several complete repetitions of a three-part programme:

1. Walk 100 m at an average speed of 6 km h^{-1}
2. Jog 200 m at an average speed of 10 km h^{-1}
3. Run 100 m at an average speed of 20 km h^{-1}

What is the athlete's average speed for the complete training session, in km h^{-1} ?

- A** 7.2
- B** 9.6
- C** 11.5
- D** 12
- E** 14.4

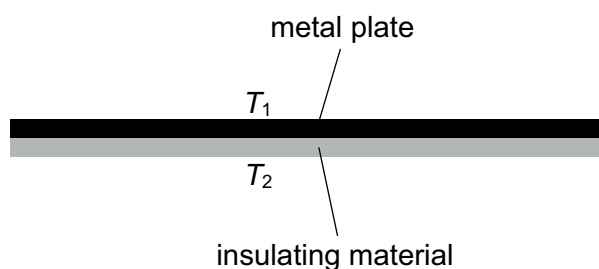
- 12** A large, flat, metal plate is coated on one side with a layer of thermally insulating material of the same thickness a as the metal plate.

The uninsulated top surface of the metal plate is maintained at a constant temperature T_1 .

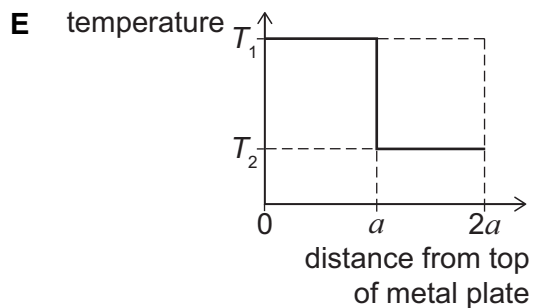
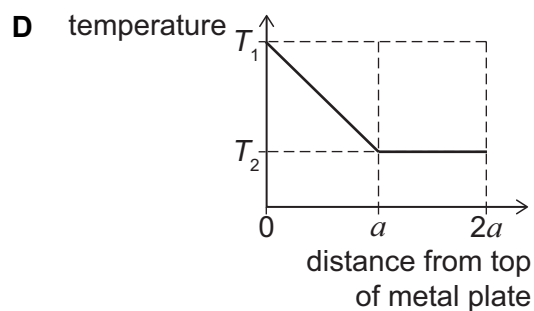
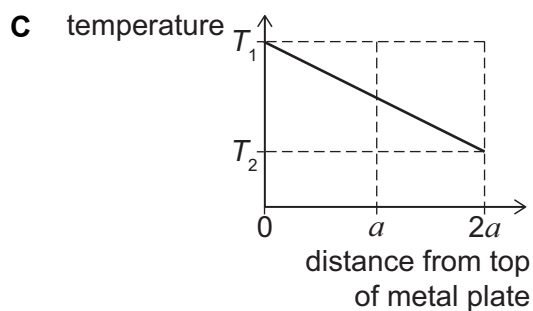
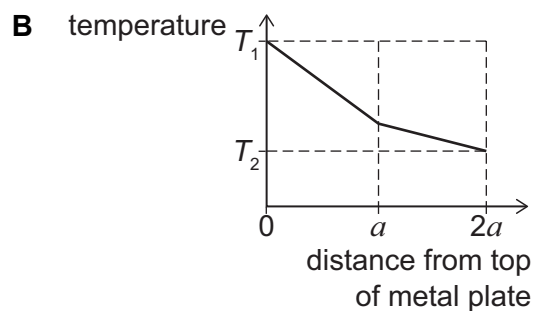
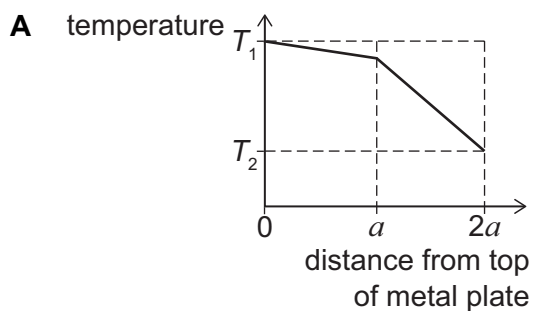
The bottom surface of the insulating material is maintained at a constant, lower temperature T_2 .

The system is in equilibrium.

The diagram shows this arrangement.



Which graph could show how the temperature varies with distance from the top surface of the metal plate to the bottom surface of the insulating material?



- 13** Two objects X and Y are similar.

The surface area of object Y is double the surface area of object X.

The volume of object Y is $7\sqrt{2} \text{ cm}^3$ more than the volume of object X.

What is the volume of object X, in cm^3 ?

A $14 - 7\sqrt{2}$

B $14 + 7\sqrt{2}$

C $\frac{42 - 7\sqrt{2}}{17}$

D $\frac{42 + 7\sqrt{2}}{17}$

E $\frac{7\sqrt{2}}{3}$

F $7\sqrt{2}$

G $4 - \sqrt{2}$

H $4 + \sqrt{2}$

- 14** The voltage output of a power station is stepped up using a transformer before the power is transmitted to a distant town. The primary coil of this transformer has 300 turns and the secondary coil has 1500 turns.

In the town, a step-down transformer reduces the voltage supplied by the transmission cables to 33 000 V for distribution within the town. The step-down transformer supplies a current of 1500 A.

The current in the transmission cables is 450 A and both transformers are ideal and 100% efficient.

What is the voltage output of the power station?

(Assume that the resistance of the transmission cables is negligible.)

A 1980 V

B 6600 V

C 22 000 V

D 110 000 V

E 550 000 V

- 15** The equation

$$\left(\frac{a \times 10^4 + 2a \times 10^3}{3 \times 10^{-1}} \right)^2 = 8 \times 10^9$$

has two solutions for a .

What is the positive difference between these two solutions?

- A** 0
 - B** $2\sqrt{5}$
 - C** $4\sqrt{5}$
 - D** $20\sqrt{5}$
 - E** $40\sqrt{5}$
 - F** $200\sqrt{5}$
- 16** A transverse wave with an amplitude of 3.0 cm travels along a stretched string. The wave has a frequency of 12 Hz and a wavelength of 0.25 m.

What is the average speed of a particle in the string as the string oscillates during a time of 2.0 s?

- A** 36 cm s^{-1}
- B** 72 cm s^{-1}
- C** 125 cm s^{-1}
- D** 144 cm s^{-1}
- E** 300 cm s^{-1}

- 17 X and Y are the end-points of a line segment.

Point P has coordinates $(-8, 5)$.

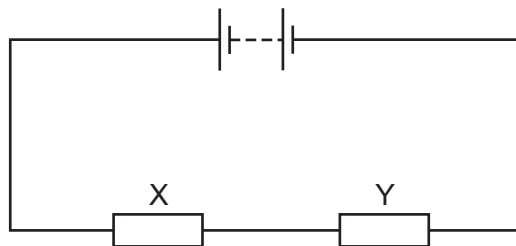
P lies on the line segment XY such that $XP:PY$ is $1:2$ and $\overrightarrow{XP} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

A point Q is such that $\overrightarrow{QY} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$

What are the coordinates of point Q ?

- A $(7, 5)$
- B $(3, 8)$
- C $(1, -12)$
- D $(-3, -10)$
- E $(-7, -7)$
- F $(-11, -4)$

- 18 A battery and two resistors X and Y are connected in series.



The power transferred by the battery is 6 W.

The resistance of X is $10\ \Omega$.

The voltage across Y is 4 V.

What is the current in the circuit?

- A $\frac{2}{5}\text{ A}$
- B $\frac{3}{5}\text{ A}$
- C $\frac{3}{4}\text{ A}$
- D 1 A
- E $\sqrt{\frac{3}{10}}\text{ A}$
- F $\sqrt{\frac{3}{5}}\text{ A}$

- 19 Find the maximum value of

$$2^{\sin x} \times 3^{-\sin x}$$

where $0^\circ \leq x \leq 360^\circ$

- A $\frac{2}{3}$
B 1
C $\frac{3}{2}$
D 2
E 3
F 6
- 20 A diver at the bottom of a lake of depth d fills a syringe with an ideal gas and seals the nozzle. The piston remains free to move. The volume of the gas in the syringe at the bottom of the lake is 90 cm^3 .

As the diver returns to the surface, the temperature of the gas does not change. At the surface of the lake the gas in the syringe is at atmospheric pressure and the volume of the gas is 720 cm^3 .

What is the volume of the gas in the syringe at a depth $\frac{d}{2}$?

- A 160 cm^3
B 180 cm^3
C 206 cm^3
D 225 cm^3
E 288 cm^3
F 315 cm^3
G 360 cm^3
H 405 cm^3

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PART B Advanced Mathematics and Advanced Physics

NOTE: questions in this part that are not covered by the ESAT content specification are indicated by a cross through the question number.

- 21 $x^2 - x - 6$ is a factor of $x^3 + ax^2 + 2x + b$, where a and b are real constants.

What is the value of $a + b$?

- A -39
- B -21
- C -3
- D $-\frac{3}{5}$
- E $\frac{3}{5}$
- F 3
- G 21
- H 39

- ~~22~~ Two pipes contain air at the same temperature and pressure.

A stationary sound wave is formed in the first pipe, which is closed at one end and open at the other end. The lowest frequency of stationary sound wave that can be formed in this pipe is 4000 Hz.

The second pipe has the same length as the first pipe, but is open at both ends.

What is the lowest frequency of stationary sound wave that can be formed in the second pipe?

- A 1000 Hz
- B 2000 Hz
- C 4000 Hz
- D 8000 Hz
- E 16 000 Hz

- 23** An arithmetic progression has first term a and common difference d .

The sum of the first 9 terms plus the sum of the first 10 terms is equal to the sum of the first 11 terms.

Which of the following is a correct expression for a in terms of d ?

A $a = -\frac{13}{4}d$

B $a = -\frac{15}{4}d$

C $a = -\frac{16}{3}d$

D $a = -\frac{19}{3}d$

E $a = -7d$

F $a = -8d$



A cannon ball of mass 2.6 kg is fired horizontally from a cannon on the top of a cliff at a speed of 90 m s^{-1} .

The height of the cliff above the horizontal ground below is 45 m.

What is the magnitude of the impulse that acts on the ball between leaving the cannon and reaching the ground?

(gravitational field strength = 10 N kg^{-1} ; assume that air resistance is negligible)

A 8.7 N s

B 13 N s

C 52 N s

D 78 N s

E 117 N s

F 234 N s

G 900 N s

- 25 The first three terms of a convergent geometric progression are:

$$2p, \quad p - 3, \quad p - 7$$

What is the sum to infinity of this progression?

- A -54
- B -27
- C $-13\frac{1}{2}$
- D -2
- E 2
- F $13\frac{1}{2}$
- G 27
- H 54

- ~~26~~ A child on a sledge is travelling straight down a snow-covered slope at a constant speed of 15 m s^{-1} .

The mass of the child and sledge together is 60 kg.

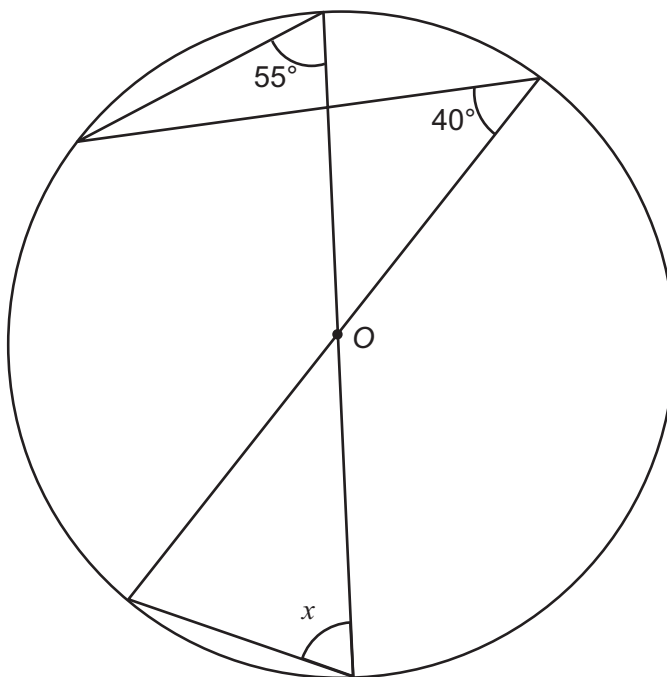
The angle of the slope to the horizontal is 30° .

What is the rate at which thermal energy is being produced due to friction forces?

(gravitational field strength = 10 N kg^{-1})

- A 450 W
- B 900 W
- C 4500 W
- D $4500\sqrt{3} \text{ W}$
- E 6750 W
- F 9000 W
- G 13 500 W
- H $9000\sqrt{3} \text{ W}$

- 27 The diagram shows a circle with centre O .



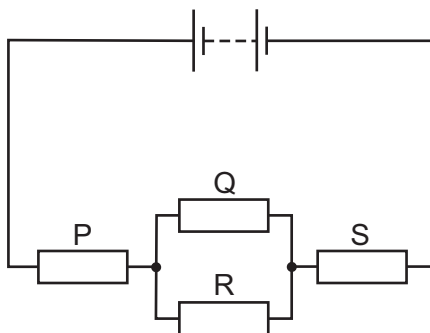
[diagram not to scale]

What is the value of x ?

- A 40°
- B 50°
- C 55°
- D 70°
- E 75°
- F 80°

- ~~26~~ The diagram shows a battery with no internal resistance and four identical resistors P, Q, R and S.

Resistor Q dissipates 4.0 W of power.



What is the total power supplied by the battery?

- A** 10 W
- B** 16 W
- C** 24 W
- D** 32 W
- E** 40 W

29 The function

$$f(x) = \sqrt{2}x^2 - 6x + 4$$

can be written in the form

$$f(x) = p(x + q)^2 + r$$

where p , q and r are constants.

What is the value of $p(r - q)$?

A 2

B 7

C $3 - \frac{\sqrt{2}}{2}$

D $3 - 7\sqrt{2}$

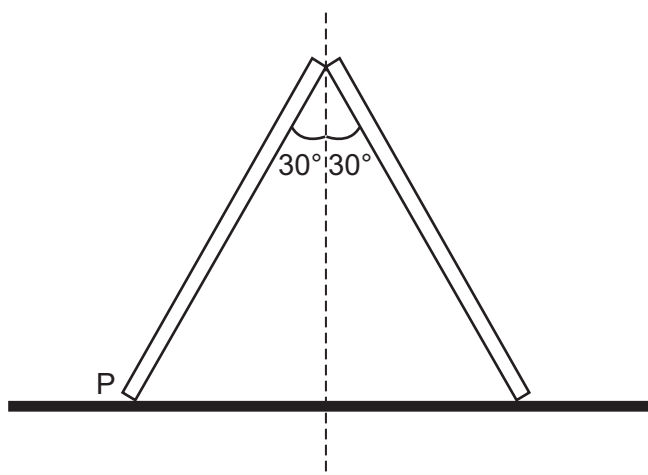
E $4 - 3\sqrt{2}$

F $4\sqrt{2} - 6$

G $4\sqrt{2} - 12$

H $7\sqrt{2} - 18$

- ✗ Two identical, uniform, thin planks, each of mass 40 kg, are propped up against one another in equilibrium on a rough surface as shown. Each plank is at an angle of 30° to the vertical.



[diagram not to scale]

What is the magnitude of the friction force at point P?

(gravitational field strength = 10 N kg^{-1})

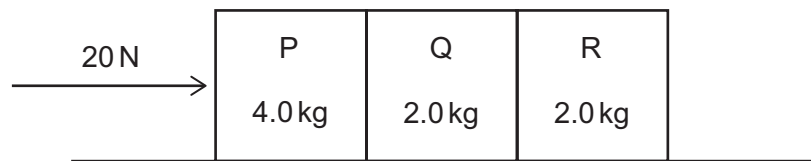
- A 100 N
- B $\frac{200}{\sqrt{3}}$ N
- C 200 N
- D $200\sqrt{3}$ N
- E $\frac{400}{\sqrt{3}}$ N
- F 400 N
- G $400\sqrt{3}$ N

31 Which of the following is equal to

$$\int_{-(3-\sqrt{5})}^{3-\sqrt{5}} \frac{x^2}{3-\sqrt{5}} dx$$

- A 0
- B $\frac{2}{3}$
- C $\frac{8}{3}$
- D 4
- E $\frac{28-12\sqrt{5}}{3}$
- F $28-12\sqrt{5}$

~~32~~ Three blocks P, Q and R are in contact on a horizontal surface as shown.



The mass of P is 4.0 kg and the masses of Q and R are each 2.0 kg.

A horizontal force of 20 N is applied to P so that all three blocks move together.

The friction force between P and the surface is 2.0 N, and the friction forces between Q and R and the surface are each 1.0 N.

What is the magnitude of the force that R exerts on Q?

- A 2.0 N
- B 5.0 N
- C 6.0 N
- D 12 N
- E 15 N
- F 17 N
- G 19 N

- 33** A function f is defined by

$$f(x) = \frac{a}{x} + \frac{b}{x^2}$$

where a and b are constants.

It is given that $f'(1) = 2$ and $f''(-1) = -2$

What is the value of $a + b$?

A -5

B -4

C $-\frac{7}{5}$

D -1

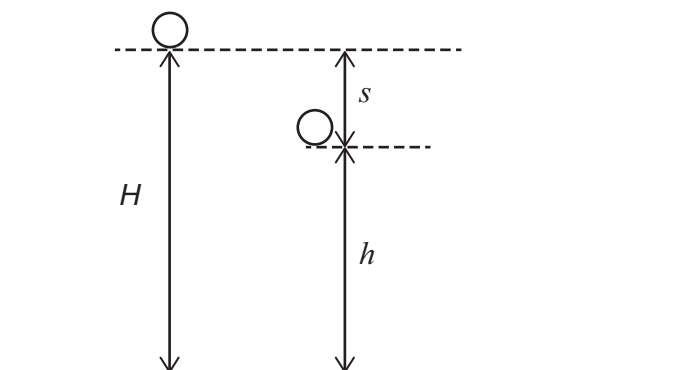
E $\frac{9}{5}$

F 2

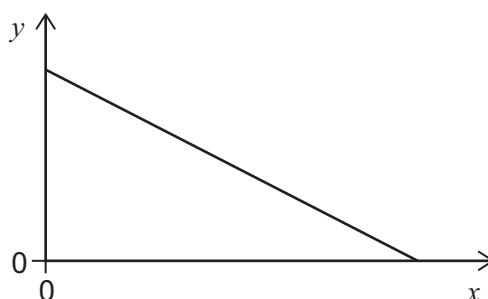
G $\frac{14}{5}$

H 3

- ✗ An object is released from rest at a height H above the ground and falls freely in a uniform gravitational field. At time t after being released, it has fallen a distance s and is at a height h above the ground, travelling at speed v .



The graph shows two quantities plotted.



Which of the rows show(s) a possible pair of quantities for the axes?

(Assume that air resistance is negligible.)

	<i>y-axis</i>	<i>x-axis</i>
1	h	t
2	kinetic energy	h
3	gravitational potential energy	s

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

35 Find the real value of x that satisfies

$$\log_3(x^2 + 3x + 2) = 2 + \log_3(x^2 + 2x)$$

- A $\frac{1}{8}$
- B $\frac{1}{7}$
- C $\frac{1}{6}$
- D $\frac{1}{2}$
- E 2
- F 7

~~36~~ A 20 V battery of negligible internal resistance is connected in series with a $20\ \Omega$ resistor and a cylindrical conductor. The current in the circuit is 0.20 A.

The cylindrical conductor is now removed, melted and re-formed into a new cylinder. After the cylinder has cooled to its original temperature, its length is 4 times greater than that of the original cylinder.

What is the resistance of the new cylinder?

- A $5.0\ \Omega$
- B $20\ \Omega$
- C $320\ \Omega$
- D $400\ \Omega$
- E $1280\ \Omega$
- F $1600\ \Omega$

37 What is the constant term in the simplified binomial expansion of $\left(\frac{2}{x} + \frac{x}{4}\right)^8$?

A $\frac{7}{2}$

B $\frac{35}{8}$

C $\frac{1}{16}$

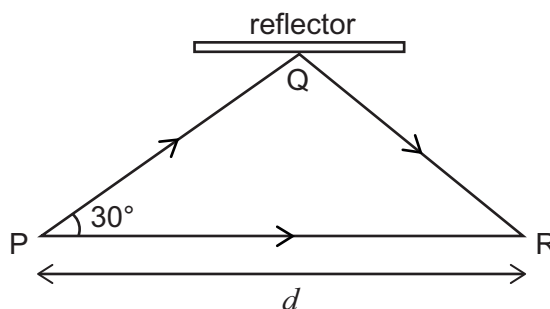
D $\frac{7}{256}$

E $\frac{35}{1024}$

F $\frac{1}{2048}$

- ✖ A sound wave can travel from a source at P to a detector at R directly or by reflecting at Q.

The angle between PR and PQ is 30° and the distance from P to R is d as shown.



There is a phase difference of π radians between the incident and reflected wave at Q.

Waves that reach R via Q are in phase with waves that reach R directly from P.

Which expression gives the greatest wavelength of sound waves for which this is true?

- A $2d\left(\frac{2}{\sqrt{3}} - 1\right)$
- B $d\left(\frac{2}{\sqrt{3}} - 1\right)$
- C $d(2 - \sqrt{3})$
- D $\frac{4d}{\sqrt{3}}$
- E $\frac{2d}{\sqrt{3}}$

39 Given that

$$x^2 + y^2 = 1$$

what is the greatest possible value of $2x + 3y$?

- A** $\frac{7}{2}$
- B** 3
- C** $\frac{5\sqrt{2}}{2}$
- D** $\sqrt{7}$
- E** $\sqrt{10}$
- F** $\frac{13\sqrt{10}}{10}$
- G** $\sqrt{13}$
- H** $\frac{12\sqrt{13}}{13}$

~~40~~ Two springs have spring constants of 200 N m^{-1} and 600 N m^{-1} , respectively.

The springs are joined in series, end-to-end, and stretched so that their combined extension is 0.80 m.

What is the total strain energy stored in the springs?

- A** 48 J
- B** 96 J
- C** 112 J
- D** 120 J
- E** 240 J
- F** 256 J
- G** 512 J

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